

Instructor: Date:

Learner: Regional Branch:

Program: Center:

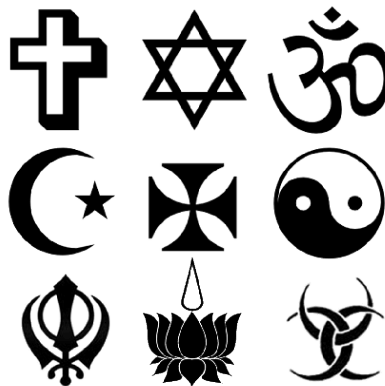
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Let's Start!

Simplifying Programing Languages



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can explain briefly what the Kernel language approach is.
- I can identify information related to formal logistics.
- I can talk about the approaches related to language semantics.

Ask your partner:

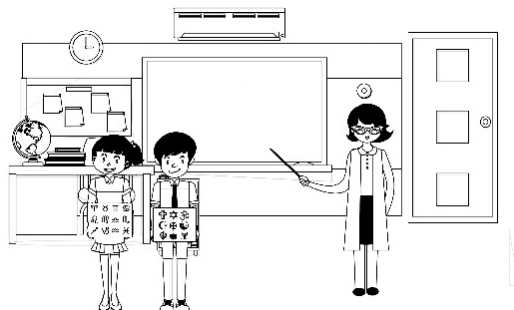
- Do you consider language approaches important? why?
- Do you know any language approach? Which one?





Let's Explore!

Sharing Experiences for Learning New Things!



Instructions: Read the text below.

Geraldine: Brandon! How nice to see you again.

Brandon: Hi, Geraldine, long time no see.

Geraldine: Yes, I know, I was a little busy the last term; I had to study hard because there was a topic I found difficult to understand.

Brandon: Really? What topic was that?

Geraldine: We studied the kernel language approach.

Brandon: Oh! The kernel language, I had to study that approach some years ago, and you are right, for me, it was also hard to grasp.

Geraldine: Yeah, but now I know that the Kernel language translates all language constructs from a core language to ease the understanding of the program functionality. And that it consists of two parts; The first one defines a simple language, which is very easy to implement and also improves efficiency. The second one defines a translation scheme from all the programming languages to the kernel language. We just need to be careful to differentiate those two aspects of the kernel language.

Brandon: There are also two kinds of translation, I remember I was always afraid to forget their names during the exams. They are the namely linguistic abstraction and the syntactic sugar; the former refers to constructs that we added to the language syntax; this extends the language and contributes to its purposes; the latter provides short-cut notations to reduce the size of the language so to improve its readability.

Geraldine: You did learn those concepts by heart. Did you learn about the approaches to the kernel language?

Brandon: Yes, I did. There are four main approaches to the kernel language semantics: a) an operational semantics approach that shows us how a statement works, b) an axiomatic semantics approach which focuses on the main relations between the inputs, c) a denotational semantics approach shows us the functionality over an abstract domain, and d) the logical semantics approach which defines a statement as a model of a logical theory.

Geraldine: I see, You didn't waste your time in class, Brandon, I feel lucky to run into you.



Let's Practice!

Don't forget the Kernel language.

Instructions: Read the text. Type in the missing words in the blanks. Use: simplified, used, and defined.

In the past, programmers _____ this approach a lot. All language constructs are terms of translations All language constructs are terms of translations _____ in a core language known as the kernel language. This language _____ the full scheme of a programming language.



Let's Practice!

Extra Kernel Language Information

Instructions: Use the words from the word bank to complete each sentence.

1. There are two kinds of language translation I'm (happy to / happy for) show you.
2. For occurring idioms we (lucky to / are lucky) to have short-cut notations to help programmers.
3. This notation is part of the language syntax, so programmers (are careful about / are careful to) define clearly using grammar rules.
4. We need to be willing (to differentiate / to differentiating) between the one that reduces the program size and the one that improves program readability



Extension Activity

There's always more to the topic

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Investigate about more characteristics and functions of the Kernel language.
- Write down your findings about Kernel language. Use the grammar covered in this unit.
- Share your ideas with your teacher and classmates.

Self-Assessment

- I investigate about Kernel language.
- I write information about Kernel language.
- I use the simple past, and adjectives with infinitives.
- I talk about Kernel language with my teacher and classmates.

Yes No

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☐ ☐
☐ ☐
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Target Words

- **core language** (noun) : principal or main language
e.g. but now I know that the Kernel language translates all language constructs from a core language to ease the understanding of the program functionality.
- **efficiency** (noun): the good use of time
e.g. The first one defines a simple language, which is very easy to implement and also improves efficiency.
- **translation scheme** (noun): an structure organized for translating
e.g. The second is important to mention because this defines a translation scheme from all the programming language to the kernel language.
- **logical** (adjective): in accordance with the reasoning
e.g. The logical semantics approach which defines a statement as a model of a logical theory.
- **program** (noun): a series of instructions for the computer to make it perform an operation.
e.g. but now I know that the Kernel language translates all language constructs from a core language to ease the understanding of the program functionality.
- **abstraction** (noun): in linguistics, a high level of analysis
e.g. They are the namely linguistic abstraction and the syntactic sugar.



Learning Points

Simple past tense in affirmative, and negative, regular vs irregular verbs

You can use the Simple Past Tense to express actions that started and finished in the past.

You use the **Simple Past** tense to talk about a complete event in the past. You can form the Simple Past tense by adding **ed, -ied** to the verb when this is regular. For irregular verbs, you have to learn its particular change.

*Example: We **designed** a new program. (regular verb)*
*They **studied** in this institution. (regular verb)*
*He **found** new software. (irregular verb)*

You need auxiliary verb **did** for the negative and interrogative form, for all the subject pronouns.

*Example: We **didn't** design a new program. (regular verb)*
***Did** they study here?*
*He **didn't** find the program.*

Notice that when you use the auxiliary did the verb takes the infinitive form.

Look at the chart of simple past

Simple Past			
	Affirmative	Negative	Questions
Regular verbs I/you/we/ they/He/she/it	I watched a new program.	We didn't send new products.	Did they get products?
Irregular verbs I/you/we/ they/He/she/it	He sold software	She didn't buy software.	Did she buy a computer?





Learning Points

Infinitives after adjectives

You can use an **infinitive** after **adjectives** to express and describe how do you feel about that action.

Remember **adjectives** are words that you use to describe the nouns.

For example:

*This exercise is very **easy***

*I am **happy** when I learn **new** things*

Infinitives almost always begin with the word **to** and after them you write the simple form of the verb which is not conjugated. It means that for the infinitive verbs you don't have to add s, or es, ed or ing at the end.

With the adjectives the structure is very simple

Subject+ verb to be+ adjective+ infinitive.

For example:

*That is **easy to explain**.*

Examples

- *I am very glad to see you again.*
- *I'm lucky to have you here to help me more with this.*
- *Those are simple translations to identify and they can have two kinds, namely linguistic abstraction and syntactic sugar.*